

FREIGHT FORWARDING & LOGISTICS ERP

Documentation

FreightFast is an enterprise-grade freight forwarding ERP — shipment requests, quotations, job files, freight documents, finance closeout, and a client portal, in one system. This page covers what it is and everything it does. For setup steps, see the [Installation Guide](#).

FRAMEWORK Next.js 16 (App Router)	LANGUAGE TypeScript, React 19	DATABASE MySQL / MariaDB + Prisma	AUTH NextAuth.js	DOCUMENTS PDFKit
AI LAYER Google Gemini (optional)				

01 — READ FIRST

What FreightFast is

A single Next.js application with a MySQL-compatible database. There's no separate backend service to stand up — the app, API routes, and background job endpoints all ship from the same codebase.

Three surfaces, one app

The build serves three distinct sign-in surfaces from the same deployment: an **operations dashboard** for company staff (quotations, shipments, finance), a **platform console** for the SaaS operator managing tenant companies, and a **client portal** your customers use to track their own shipments and documents. Routing between them is host-based, configured during installation.

Multi-tenant by design

Every record belongs to exactly one company, and within a company, to exactly one branch. If you're installing this for a single freight forwarder, you'll use one company with one or more branches; the platform console exists for cases where you're operating this as a SaaS across multiple client companies.

Role-based from the ground up

Seven predefined roles — Company Admin, Operations Manager, Documentation Officer, Accounts Officer, Sales Executive, and two more — gate every module, action, and record. Permissions are enforced server-side on every request, not just hidden in the UI.

02 — WHAT'S INSIDE

Core features

Eighteen operational modules covering the full shipment lifecycle, from first inquiry to closed invoice.

Control Tower Dashboard
Real-time KPIs, shipment visibility, workflow blocker detection, revenue & profit analytics.

Customer & CRM Management
Customer database, company profiles, contacts, billing details, shipment history.

Shipment Request Management
Inquiry intake, cargo capture, Port-to-Port / Door-to-Door scope, auto-filled ETD/ETA.

Quotation Management
Freight & local charge estimation, buy/sell rates, profit visibility, approval workflow.

Job File Management
10-tab workspace: Overview, Workflow, Commercial, Operations, Documents, Freight Docs, Finance, Portal, Tasks, Timeline.

Document Compliance Engine
Mode/type-aware document requirements, required-document gates, finance close blockers.

Freight Document Generator
PDF generation for HBL, HAWB, sea/air manifests, delivery orders, debit notes, arrival notices.

Finance & Closeout
Invoicing, vendor bills, margin audit, financial lock, audit-safe closeouts.

Client Portal

Role-Based Access Control

OTP login, forced first-time password reset, milestone tracking, document access.

Seven predefined roles, fine-grained permissions, strict company isolation.

Multi-Branch Management
Every record owned by exactly one branch; staff scoped to the branches they belong to.

Automated Status Notifications
In-app alerts on invoice/payment/POD events, plus an idempotent email/WhatsApp outbox.

Advanced Global Search
One search box across customers, shipments, quotations, invoices, vendors, and documents.

Saved Filters & Views
Private, named filter presets on list pages, with a per-user default view.

Customizable Dashboard
Reorder and show/hide dashboard sections per user, with a one-click reset.

Import & Export Data
CSV export across eight entities; a guided import wizard for customers and vendors.

Recurring Task Management
Daily/weekly/monthly task templates, generated by an idempotent self-hosted cron.

Custom Document Templates
Branded, versioned layouts for quotations, invoices, HBL, HAWB, debit notes, manifests.

Approval Workflow System
Threshold-based sign-off chains for vendor bills and payments, up to five approver steps.

03 — OPTIONAL LAYER

AI-assisted tools

Gated behind a platform kill-switch, a per-company toggle, and a daily request cap — off by default until a Gemini API key is configured. See the Installation Guide's environment reference for `AI_ENABLED`.

TOOL	WHAT IT DOES	POWERED BY
Shipment Assistant	Chat-style Q&A over a shipment's status, tasks, and open issues.	Gemini

Document Reader	Extracts shipment and document fields from an uploaded file.	Gemini
Document Checker	Flags missing, incorrect, or inconsistent fields on a freight document.	Gemini
Document Cross-Check	Compares multiple documents on a shipment and flags mismatches.	Gemini
Quotation Generator	Drafts charge lines from customer, shipment, and pricing context.	Gemini
Email Generator	Drafts a customer-ready email from shipment and operational context.	Gemini
Shipment Summary	Turns a full job file into a short operational summary.	Gemini
Report Insights	Explains notable trends and changes on financial reports.	Gemini
Smart Search	Natural-language mode for the global search box.	Gemini
Customer Insights	Summarizes a customer's history, activity, and opportunities.	Gemini
Delay Alerts	Flags shipments at risk of delay from ETA and workflow data.	Rule-based
Profit Analysis	Highlights low-margin shipments against historical averages.	Rule-based
Task Suggestions	Recommends next actions from open exceptions and pending work.	Rule-based
Shipment Health Score	Scores a shipment from status, milestones, documents, and conditions.	Rule-based
Exception Radar	Surfaces shipments, documents, or workflows that need attention.	Rule-based aggregator

 "Rule-based" tools don't call an external model — they're deterministic logic over your own data. They still route through the same permission and company-toggle gate as the Gemini-backed tools, so access control stays consistent either way.

For setup and configuration issues, start with the [Installation Guide](#) — it covers environment variables, default accounts, and a troubleshooting section for the most common setup problems. Keep both documents alongside your copy of the source for reference during future upgrades.